



IPv6 Essentials

Use this when: you read, shorten or configure an IPv6 address, or need to know which type it is and how it was assigned.

CCNA 200-301 · topics 1.8, 1.9
Sheet v1.0 · Oct 2026

🔧 Anatomy of an address: 128 bits, 8 groups

Example: 2001:db8:acad:1::10/64

2001 group 1	0db8 group 2	acad group 3	0001 group 4	0000 group 5	0000 group 6	0000 group 7	0010 group 8
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Global routing prefix · 48 bits
Given by your ISP or RIR

Subnet · 16
65,536 LANs

Interface ID · 64 bits
Host: SLAAC, EUI-64 or manual

Shorten (RFC 5952): drop leading zeros, then `::` replaces the longest all-zero run (once only). Lowercase hex.
2001:0db8:acad:0001:0000:0000:0000:0010 becomes 2001:db8:acad:1::10

🔧 Address types

No broadcast in IPv6

2000::/3	Global unicast	Routable on the internet
fe80::/10	Link-local	One link only; every interface has one
fc00::/7	Unique local	Private; in practice <code>fd00::/8</code>
ff00::/8	Multicast	One to many; replaces broadcast
::1/128	Loopback	This host (<code>::/128</code> = no address yet)
2001:db8::/32	Documentation	Use in examples, never on a live network
MULTICAST	ff02::1 all nodes · ff02::2 all routers · ff02::1:ffXX:XXXX solicited-node	



Neighbor Discovery

Replaces ARP · ICMPv6

133	RS	Host asks for routers
134	RA	Router announces prefix, gateway
135	NS	"Who has this address?"
136	NA	"I do, here is my MAC"
137	Redirect	Router names a better first hop
NS/NA = ARP, plus finding duplicate addresses		
HOP LIMIT 255 on every ND message, so off-link senders are ignored		

⚙️ How a host gets its address

SLAAC	Host builds its ID from the RA prefix. Needs a /64.
DHCPv6	Stateless adds DNS; stateful assigns the address
EUI-64	Flip bit 7 of the MAC, insert <code>fffe</code> in the middle
EXAMPLE	MAC 00:00:5e:00:53:01 becomes 0200:5eff:fe00:5301



IPv4 vs IPv6

Address	32 bits, dotted decimal · 128 bits , hex
Header	20+ bytes, checksum · 40 bytes fixed , no checksum
Fragmenting	Routers can · Source only (min link MTU 1280)
NAT	Common · Not needed ; use a firewall

🔧 Configure and check

ipv6 unicast-routing	Routing on
ipv6 address 2001:db8:acad:1::1/64	Static address
ipv6 address 2001:db8:acad:1::/64 eui-64	EUI-64 ID
ipv6 route ::/0 2001:db8:acad:1::2	Default route
show ipv6 interface brief	Up/up? IPs
show ipv6 route · show ipv6 neighbors	Routes, ND
ping -6 · ipconfig · ip -6 addr	Win, Win, Linux



Secure it

⚠️ **DANGER** A rogue **Router Advertisement** can hijack the LAN. Enable **RA Guard** on access ports.

⚠️ **CAUTION** Never block all ICMPv6: types 133 to 136 and Packet Too Big must pass.

⚠️ **CAUTION** IPv6 is on by default. Firewall rules must cover it too.



REMEMBER

- /64 per LAN, /48 or /56 per site
- fe80:: is on every interface
- :: only **once** per address